Examiner only

8. (a) Use the information under **Figure 1** to answer the following question.
The wave speed of ultrasound in body tissue is 1540 m/s.

Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of the ultrasound used in a 2 MHz scan. [3]

wavelength = ... m

(b) Use the information in **Figure 4** to answer the following questions.

(i) Select the most suitable radioisotope of iodine from the table in **Figure 4** that would be used as a tracer. Explain your choice. [2]

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Examiner only

(ii) The photograph shows a patient undergoing external beam radiation therapy for a brain tumour.

Select the most suitable radioisotope from the table in **Figure 4** that would be used for this treatment. Explain your choice. [3]

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(c) Refer to **Figures 4 and 6** in the Resource Folder to answer the following question.

State which of the radioisotopes **R1**, **R2**, **R3**, **R4**, **R5** or **R6**, is lutetium-177. Explain your reasoning. [3]

Radioisotope .... is lutetium-177.

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